

Professor Recommendation System Using Multi-Source Sentiment Analysis

Problem Statement

Students often rely on informal sources when deciding which professors to take for a course. Platforms such as RateMyProfessors and online discussion forums like Reddit contain large amounts of feedback about instructors, including comments about teaching style, grading fairness, workload, and overall course experience. However, this information is scattered across multiple platforms and presented in an unstructured format, making it difficult for students to systematically evaluate instructor quality.

The goal of this project is to develop a data-driven professor recommendation system that aggregates publicly available online discussions and reviews about professors and applies sentiment analysis to extract meaningful insights. By collecting and analyzing text data from platforms such as RateMyProfessors and Reddit, the system will measure sentiment and identify key themes related to instructor performance. The final platform will allow users to search for a professor and view aggregated sentiment insights and recommendation scores based on student feedback.

Strategic Aspects

This project focuses on building a complete data pipeline and analytics platform that integrates data acquisition, database design, natural language processing, and web application development.

First, publicly available text data related to professors will be collected from online sources such as RateMyProfessors and Reddit discussions. These sources contain large volumes of qualitative feedback that reflect students' experiences in various courses.

Next, the collected text will be cleaned and standardized using Python. Data preprocessing will include removing duplicates, filtering irrelevant content, normalizing text formats, and extracting useful features such as professor names and course identifiers.

The processed data will be stored in a PostgreSQL relational database, which will organize information about professors, reviews, sentiment scores, and data sources. A structured database allows the system to efficiently query and analyze aggregated review data.

Natural language processing techniques will then be applied to the review text. Sentiment analysis models will evaluate whether comments express positive, neutral, or negative opinions about instructors. Additional keyword analysis will identify common themes such as clarity, fairness, workload, and helpfulness.

Finally, the processed data and sentiment results will be served through a Django web application that allows users to search for professors, explore sentiment summaries, and view recommendation scores derived from aggregated student feedback.

Course Relation to Lectures and Discussions

This project demonstrates several key concepts related to data management, database design, and data analytics workflows.

First, the project involves acquiring and integrating external datasets from online platforms, demonstrating data acquisition and multi-source data integration.

Second, the project requires designing and implementing a relational database schema using PostgreSQL. Tables will represent professors, reviews, data sources, and sentiment analysis results. This connects directly to relational modeling, schema design, and efficient storage practices.

Third, the project includes preprocessing and transforming raw textual data into structured information using Python libraries such as Pandas. This reflects ETL workflows and preparing data for analysis.

Finally, sentiment analysis and data visualization techniques will be used to generate insights from the collected data, demonstrating how database systems and analytical methods can work together to produce decision-support tools.

Novelty and Importance

While platforms like RateMyProfessors provide ratings for instructors, they often present feedback in fragmented formats. Online forums such as Reddit also contain valuable discussions about professors and courses but lack structured aggregation and analysis.

This project transforms unstructured online feedback into structured, analyzable information. By combining sentiment analysis with multiple public data sources, the system can identify broader trends and generate more objective indicators of teaching quality and student satisfaction.

Additionally, the project demonstrates how natural language processing can be applied to real-world problems involving large volumes of user-generated text. Educational data analytics is becoming increasingly important for understanding student experiences and improving academic decision-making.

Data Description

The project will collect publicly available text data related to professors from several online platforms. Potential sources include RateMyProfessors review pages and Reddit discussions related to university courses and instructors.

Each data record will include information such as professor name, department or course reference (if available), review text, source platform, and timestamp when available.

After preprocessing, additional derived attributes will be generated, including sentiment score (positive, neutral, negative), keyword indicators related to teaching traits, aggregated recommendation score for each professor, and frequency of commonly mentioned themes such as clarity, fairness, workload, and helpfulness.

Methodology and Plan

The project will be implemented in several stages using a specific Python, Django, and PostgreSQL workflow.

First, I will collect public discussion data from Reddit using the PRAW API and from professor review pages using requests and BeautifulSoup, limited to publicly accessible information. The collection scripts will extract professor names, course references when available, review text, post timestamps, source platform, and URLs. I will save the raw scraped output as CSV or JSON files so the ingestion process is reproducible.

Second, I will build a preprocessing pipeline in Python using Pandas, regular expressions, and spaCy. This stage will clean HTML artifacts, remove duplicate posts, standardize professor name variations, normalize course codes, and filter irrelevant or extremely short comments. I will also create validation checks to flag missing professor names, malformed records, and duplicate source URLs before loading data into the database.

Third, I will design a PostgreSQL schema with tables such as Professors, Courses, Sources, Reviews, and SentimentResults. I will use Django models to implement this schema, define foreign-key relationships, and enforce constraints for referential integrity. I will then create a Django management command that loads the cleaned CSV or JSON data into PostgreSQL so the ETL process can be rerun whenever new data is collected.

Fourth, I will perform sentiment analysis using NLTK's VADER model for baseline sentiment scoring and spaCy for tokenization and keyword extraction. For each review, I will generate a sentiment label and sentiment score, and I will also extract keywords related to specific teaching traits such as clarity, fairness, workload, and helpfulness. These results will be written back into PostgreSQL so that the application can query both the raw review text and the processed analytics.

Fifth, I will compute an overall professor recommendation score using aggregated sentiment metrics in Python. This score will combine average sentiment, frequency of

positive and negative keywords, and review volume so that recommendations are based on both quality and consistency of feedback. I will use SQL queries and Pandas summaries to calculate these aggregates and compare professors within the same department or course area.

Finally, I will build a Django web application that allows users to search for professors, compare professors, and view sentiment summaries. The frontend pages will display review counts, average sentiment, recommendation scores, and charts built with Plotly or Matplotlib. The final system will therefore include a reproducible ingestion pipeline, a normalized PostgreSQL database, NLP-based review analysis, and a working web interface for exploring professor recommendations.

Tools and Technologies

Python for data collection, preprocessing, ETL scripting, and recommendation-score calculations.

Django for backend application development, database integration through models, and professor search and comparison views.

PostgreSQL for relational data storage, query execution, constraints, and aggregation of review and sentiment data.

Pandas for tabular cleaning, transformation, validation checks, and summary statistics.

PRAW, requests, and BeautifulSoup for collecting public online text data from Reddit and professor review pages.

NLTK VADER and spaCy for sentiment analysis, tokenization, text normalization, and keyword extraction.

Plotly or Matplotlib for visualizations such as sentiment distributions, recommendation rankings, and review-count charts.

Expected Outcome

The final result will be a professor recommendation platform that aggregates publicly available student feedback from multiple online sources and analyzes sentiment using natural language processing techniques. The platform will allow users to search for instructors and explore aggregated sentiment insights and recommendation scores.

This project demonstrates how database systems, data pipelines, and machine learning techniques can be combined to transform unstructured online feedback into meaningful insights that support student decision-making.